Lesson 7: Camera Models & Camera Control

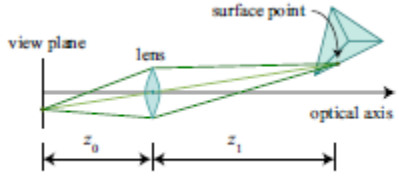
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7.1 Thin Lens & Pinhole Camera Model

7.1.1 Thin Lens

Most modern cameras use a lens to focus light onto the view plane (i.e., the sensory surface). This is done so that one can capture enough light in a sufficiently short period of time that the objects do not move appreciably, and the image is bright enough to show significant detail over a wide range of intensities and contrasts.

Lens models can be quite complex, especially for compound lens found in most cameras. Here we consider perhaps the simplest case, known widely as the thin lens model. In the thin lens model, rays of light emitted from a point travel along paths through the lens, converging at a point behind the lens. The key quantity governing this behavior is called the **focal length** of the lens. The **focal length**, |f|, can be defined as distance behind the lens to which rays from an infinitely distant source converge in focus.

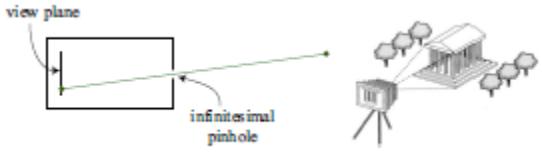


More generally, for the thin lens model, if z_1 is the distance from the center of the lens (i.e., the nodal point) to a surface point on an object, then for a focal length $|f|$, the rays from that surface point will be in focus at a distance z_0 behind the lens center, where z_1 and z_0 satisfy the thin lens equation:

$$\frac{1}{|f|} = \frac{1}{z_0} + \frac{1}{z_1} \qquad (1)$$

7.1.2 Pinhole Camera Model

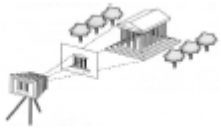
A **pinhole** camera is an idealization of the thin lens as aperture shrinks to zero.



Light from a point travels along a single straight path through a pinhole onto the view plane. The object is imaged upside-down on the image plane.

Remember:

We use a right-handed coordinate system for the camera, with the x-axis as the horizontal direction and the y-axis as the vertical direction. This means that the optical axis (gaze direction) is the negative z-axis.



For the remainder of these notes, we will consider this camera model, as it is somewhat easier to think about, and also consistent with the model used by OpenGL.

Aside:
The earliest cameras were room-sized pinhole cameras, called *camera obscuras*. You would walk in the room and see an upside-down projection of the outside world on the far wall. The word *camera* is Latin for "room;" *camera obscura* means "dark room."



18th-century camera obscuras. The camera on the right uses a mirror in the roof to project images of the world onto the table, and viewers may rotate the mirror.

7.2 Camera Projections

A **3D projection** or **graphical projection** maps points in three-dimensions onto a two-dimensional plane. As graphics are usually displayed on two-dimensional media such as paper and computer monitors, these projections are widely used, especially in engineering drawing, drafting, and computer graphics. Projections may be calculated mathematically or by various geometrical or optical techniques.

Projection is achieved by the use of imaginary "projectors"; the projected, mental image becomes the technician's vision of the desired, finished picture. Methods provide a uniform imaging procedure among people trained in technical graphics (mechanical drawing, computer aided design, etc.). By following a method, the technician may produce the envisioned picture on a planar surface such as drawing paper.

There are two graphical projection categories, each with its own method:

- parallel projection
- perspective projection



Parallel projection

Parallel projection corresponds to a perspective projection with a hypothetical viewpoint; i.e. one where the camera lies an infinite distance away from the object and has an infinite focal length, or "zoom".

In parallel projection, the lines of sight from the object to the projection plane are parallel to each other. Thus, lines that are parallel in three-dimensional space remain parallel in the two-dimensional projected image. Parallel projection also corresponds to a perspective projection with an infinite focal length (the distance from a camera's lens and focal point), or "zoom".

Perspective projection

Perspective projection is a linear projection where three dimensional objects are projected on a *picture plane*. This has the effect that distant objects appear smaller than nearer objects.

It also means that lines which are parallel in nature (that is, meet at the point at infinity) appear to intersect in the projected image, for example if railways are pictured with perspective projection, they appear to converge towards a single point, called the vanishing point. Photographic lenses and the human eye work in the same way, therefore perspective projection looks most realistic.Perspective

projection is usually categorized into *one-point*, *two-point* and *three-point perspective*, depending on the orientation of the projection plane towards the axes of the depicted object.

Graphical projection methods rely on the duality between lines and points, whereby two straight lines determine a point while two points determine a straight line. The orthogonal projection of the eye point onto the picture plane is called the *principal vanishing point* (P.P. in the scheme on the left, from the Italian term *punto principale*, coined during the renaissance).

Two relevant points of a line are:

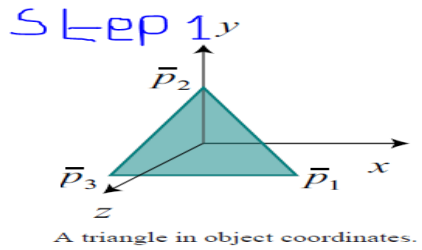
- its intersection with the picture plane, and
- its vanishing point, found at the intersection between the parallel line from the eye point and the picture plane.

The principal vanishing point is the vanishing point of all horizontal lines perpendicular to the picture plane.

7.3 Projecting a Triangle

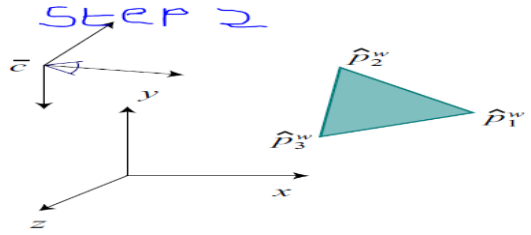
Let's review the steps necessary to project a triangle from object space to the image plane.

1. A triangle is given as three vertices in an object-based coordinate frame: $\bar{p}_1, \bar{p}_2, \bar{p}_3$

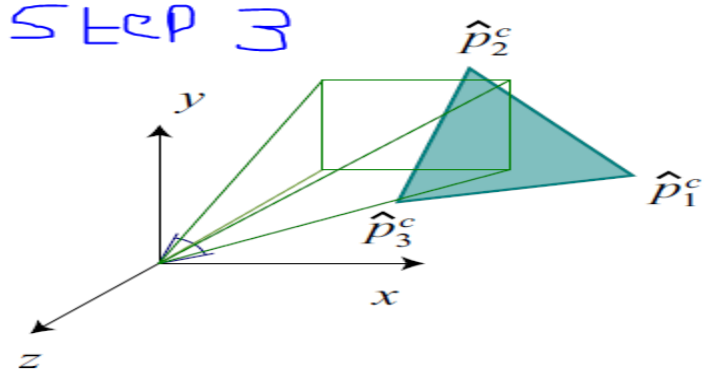


A triangle in object coordinates.

2. Transform to world coordinates based on the object's transformation: $\hat{p}_{w1}$, $\hat{p}_{w2}$, $\hat{p}_{w3}$, where $\hat{p}_{wi} = \hat{M}_{ow}\ \hat{p}_{oi}$



3. Transform from world to camera coordinates: $\hat{p}_{ci} = \hat{M}_{wc}\ \hat{p}_{wi}$



The triangle projected from world to camera coordinates.

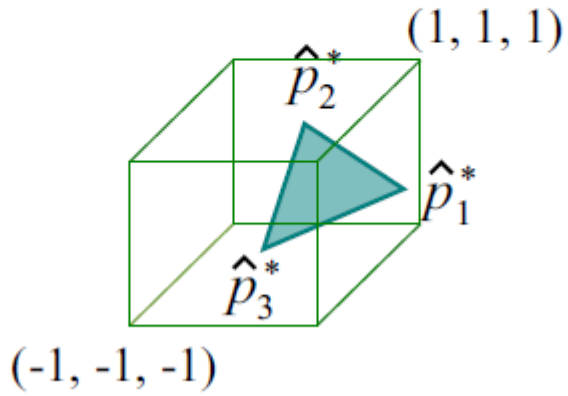
4. Homogeneous perspective transformation: $\hat{x}_{*i} = \hat{M}_p \hat{p}_{ci}$, where

$$\hat{M}_p = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & a & b \\ 0 & 0 & 1/f & 0 \end{bmatrix}, \text{ so } \hat{x}_i^* = \begin{bmatrix} p_x^c \\ p_y^c \\ ap_z^c + b \\ \frac{p_z^c}{f} \end{bmatrix}.$$

5. Divide by the last component:

Step 5

$$\begin{bmatrix} x^* \\ y^* \\ z^* \end{bmatrix} = f \begin{bmatrix} \frac{p_x^c}{p_z^c} \\ \frac{p_y^c}{p_z^c} \\ \frac{ap_z^c + b}{p_z^c} \end{bmatrix}.$$



The triangle in normalized device coordinates after perspective division.

7.4 Camera Projections in OpenGL

OpenGL's model view matrix is used to transform a point from object or world space to camera space. In addition to this, a *projection matrix* is provided to perform the homogeneous perspective transformation from camera coordinates to *clip coordinates* before performing perspective division. After selecting the projection matrix, the glFrustum function is used to specify a viewing volume, assuming the camera is at the origin:

```
glMatrixMode(GL_PROJECTION); glLoadIdentity();
glFrustum(left, right, bottom, top, near, far);
```

For orthographic projection, glOrtho can be used instead:

```
glOrtho(left, right, bottom, top, near, far);
```

The GLU library provides a function to simplify specifying a perspective projection viewing frustum:

gluPerspective(fieldOfView, aspectRatio, near, far);

The field of view is specified in degrees about the *x*-axis, so it gives the vertical visible angle. The aspect ratio should usually be the viewport width over its height, to determine the horizontal field of view.

Lesson 7 Discussion Session

1. How do you create an environment map to capture lighting effects from a real-world outdoor scene? Assume only tools that you may have in your home (e.g., a camera).
2. Discuss advantages and disadvantages of thin lenses and pinhole camera.
3. Discuss each of the following projection:
    A. Orthographic projection
    B. Multiview projection
    C. Axonometric projection
    D. Isometric projection
    E. Dimetric projection
    F. Trimetric projection
    G. Oblique projection